**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 076 · Lesson 076 · Algorithm Choice: Time, Space and Preconditions

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What must binary search assume?

Q2. Define auxiliary space.

Q3. Why can preprocessing change a choice?

Q4. Binary search in [1,3,5,7,9,11,13,15,17] for 15: give midpoint indexes.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

precondition — Input requirement for correctness.

preprocessing — Preparing data before main operation.

amortise — Spread one-off cost across tasks.

## New learning

Understand — Choose an algorithm based on input properties, particularly whether data are already sorted.

Trace — Compare running time AND auxiliary memory: linear scan O(n) time, O(1) extra; iterative binary search O(log n), O(1) extra on sorted data.

Explain — Sorting first takes time; consider that one-time cost and the number of future searches.

## Worked example

Choose a method to find 7 in unsorted [4,7,2].

1. Check the working — Linear search: inspect4 then7, returning index1.

# We Do · Practise with guidance

Find 12 in sorted [2,5,7,9,12,16,20] using binary search.

```
mid3=9, mid5=16, mid4=12, return index4.
```

Check the working

mid3=9, mid5=16, mid4=12, return index4.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Choose linear/binary for unsorted [8,2,5,9], target5. [4]

### Question 2

Trace binary search for 12 in [2,5,7,9,12,16,20]. [6]

### Question 3

Compare one search in short unsorted list versus many searches in large fixed dataset. [6]

# Exit Ticket · Answer before checking

Binary search must always be preferred even for one lookup in unsorted data.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Data are sorted.

Q2. Extra working memory beyond input storage.

Q3. A one-time sort has cost.

Q4. mid4=9, mid6=13, mid7=15; return index7.

### You Do pages · Independent questions

Q1. Linear [2]; binary requires sorted data [2].

Q2. mid3=9 [2]; low4 mid5=16 [2]; high4 mid4=12, index4 [2].

Q3. Linear single search avoids sort [2]; sort once O(n log n) [2], then binary O(log n) repeatedly [2].

### Exit Ticket · Model answer

False. Sorting first may cost more.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**